Huawei Access Node BAS-ACC-73 Basavanagudi POP Routing Protocol Flap Resolution SOP

Node: BAS-ACC-73 | **Vendor:** Huawei | **Severity:** Minor | **Alarm Code:** ALARM_9597

Description

This Standard Operating Procedure (SOP) addresses the failure domain of Routing Protocol Flap (BGP route churn, OSPF adjacency loss, ISIS LSP flooding) on Huawei Access Node BAS-ACC-73 located at Basavanagudi POP. The Routing Protocol Flap is a condition where the routing protocol adjacency or adjacency loss results in frequent and rapid changes in the routing table, leading to sub-optimal routing decisions and increased network instability. This is typically caused by BGP route churn, OSPF adjacency loss, or ISIS LSP flooding due to misconfiguration, hardware issues, or software bugs on the Huawei Access Node. The Huawei Access Node plays a critical role in routing protocol convergence and stability by maintaining routing information and forwarding packets. However, in the event of a Routing Protocol Flap, it is essential to quickly identify and resolve the root cause to prevent network-wide disruptions.

Pre-requisite Checks

- Verify the Basavanagudi POP is accessible and dispatch remote hands as needed.

- Confirm the redundant link status for Node BAS-ACC-73 to ensure high availability.

- Check for any recent software upgrades or configuration changes that may have contributed to the Routing Protocol Flap.

Potential Root Causes

Cause	Probability
BGP Route Churn due to misconfiguration or software bug	High
OSPF Adjacency Loss due to hardware issues or configuration error	Medium
ISIS LSP Flooding due to software bug or configuration mismatch	Low

Diagnostic Commands & Expected Outputs

```
Command: display bgp routing-table, Expected Output: Display the current BGP routing table for verification of BGP route churn., Description: This command will help verify if there is any BGP route churn by displaying the current BGP routing table.

Command: display ospf neighbor, Expected Output: Display the OSPF neighbor status for verification of OSPF adjacency loss., Description: This command will help verify if there is any OSPF adjacency loss by displaying the OSPF neighbor status.

Command: display isis lsp, Expected Output: Display the ISIS LSP status for verification of ISIS LSP flooding., Description: This
```

```
command will help verify if there is any ISIS LSP flooding by
displaying the ISIS LSP status.
```

Resolution Steps

Step 1: Step: Verify and correct BGP configuration, Command: display bgp configuration, Description: Verify the BGP configuration and correct any misconfiguration that may have contributed to the BGP route churn.

Step 2: Step: Verify and correct OSPF configuration, Command: display ospf configuration, Description: Verify the OSPF configuration and correct any misconfiguration that may have contributed to the OSPF adjacency loss.

Step 3: Step: Verify and correct ISIS configuration, Command: display isis configuration, Description: Verify the ISIS configuration and correct any misconfiguration that may have contributed to the ISIS LSP flooding.

Post-Check Verification

```
Command: display bgp routing-table, Expected Output: Verify the BGP
routing table for confirmation of resolution., Description: Verify the
BGP routing table for confirmation of resolution.

Command: display ospf neighbor, Expected Output: Verify the OSPF
neighbor status for confirmation of resolution., Description: Verify
the OSPF neighbor status for confirmation of resolution.

Command: display isis lsp, Expected Output: Verify the ISIS LSP status
for confirmation of resolution., Description: Verify the ISIS LSP
status for confirmation of resolution.
```

Roll-back Procedures

- Step: Restore the original BGP configuration, Command: undo bgp configuration, Description: Restore the original BGP configuration in the event of unsuccessful resolution.

- Step: Restore the original OSPF configuration, Command: undo ospf configuration, Description: Restore the original OSPF configuration in the event of unsuccessful resolution.

- Step: Restore the original ISIS configuration, Command: undo isis configuration, Description: Restore the original ISIS configuration in the event of unsuccessful resolution.